

Probability Density Function

```
def get_pdf_probability(dataset, startrange, endrange):
    from matplotlib import pyplot
    from scipy.stats import norm
    import seaborn as sns
    ax = sns.distplot(dataset, kde=True, kde_kws={'color': 'blue'}, color='Green')
    pyplot.axvline(startrange, color='Red')
    pyplot.axvline(endrange, color='Red')
    # generate a sample
    sample = dataset
    # calculate parameters
    sample_mean = sample.mean()
    sample_std = sample.std()
    print('Mean=%.3f, Standard Deviation=%.3f' % (sample_mean, sample_std))
    # define the distribution
    dist = norm(sample_mean, sample_std)

    # sample probabilities for a range of outcomes
    values = [value for value in range(startrange, endrange)]
    probabilities = [dist.pdf(value) for value in values]
    prob = sum(probabilities)
    print("The area between range({}, {}): {}".format(startrange, endrange, sum(probabilities)))
    return prob
```

This Python code defines a function, `get_pdf_probability`, designed to visualize a dataset's distribution and calculate the approximate probability of a value falling within a specific range.

Essential Information

The function takes a **dataset** and a **numerical range** (startrange to endrange) as inputs and performs the following:

- **Visualization:** It uses the seaborn and matplotlib libraries to plot a histogram and Kernel Density Estimate (KDE) of the data. It also draws red vertical lines to mark the specified range.
- **Statistical Modelling:** It assumes the data follows a **Normal (Gaussian) Distribution**. It calculates the mean and standard deviation of the input dataset to define this distribution.
- **Probability Calculation:** It approximates the "area under the curve" (probability) by summing the Probability Density Function (PDF) values for every integer within the user-defined range.
- **Output:** It prints the calculated mean, standard deviation, and the final probability, which it then returns.

Summary Table

Component	Description
Libraries Used	matplotlib.pyplot, scipy.stats.norm, seaborn
Primary Goal	Visualize data and estimate range probability.
Assumption	The dataset is normally distributed.
Key Output	A float representing the summed PDF values (prob).

$$Z \text{ Score} = (x - \bar{x})/\sigma$$

Where, x = Standardized random variable. $\bar{x}$ = Mean. σ = Standard deviation.

Z-score normalization is a statistical method that transforms data points into standardized scores by expressing them in terms of standard deviations from the mean. This technique is crucial for comparing data across different scales and distributions, particularly in time-series analysis and financial applications.